Abstract #00897 ISMRM ABSTRACT

Deep Residual Learning for Artifact Suppression in Simultaneous Sodium MRI–EEG Acquisition

Primary: Acquisition & Reconstruction - Artifacts and Correction Strategies **Secondary:** Analysis Methods - Classification and Prediction **Keywords:** ARTIFACTS DEEP LEARNING DENOISING SODIUM MRI-EEG EEG U-NET

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Impact

This work evaluates EEG artifact removal strategies during sodium MRI. With sufficient training data, supervised learning demonstrates superior performance in mitigating MRI-induced interference compared with conventional ICA, enabling cleaner EEG signals for simultaneous sodium MRI–EEG acquisitions.

Synopsis

Motivation: Simultaneous sodium MRI–EEG acquisition offers complementary spatial–temporal information but suffers from severe gradient-induced artifacts that obscure neural signals.

Goals: To develop and systematically compare artifact-removal methods capable of recovering physiologically meaningful EEG and ECG signals.

Approach: Three denoising strategies were evaluated: (1) Independent Component Analysis, (2) Residual U-Net, and (3) Variational Autoencoder—using EEG power spectra and ECG-based RMSE and Pearson correlation as quantitative benchmarks.

Results: The comparison reveals distinct interactions between deep learning architectures and low-SNR data, identifying the Residual U-Net as the most effective model for suppressing MRI interference while preserving physiological signal integrity.

Introduction

Simultaneous MRI–EEG combines the millisecond temporal precision of electroencephalography (EEG) with MRI's spatial resolution, enabling direct observation of neural dynamics, neurovascular coupling, and metabolic processes. Yet, EEG acquired inside the MRI scanner is heavily contaminated by gradient-induced artifacts whose amplitudes can exceed neural signals by up to two orders of magnitude (Figure 1). These artifacts obscure genuine electrophysiologic activity and remain a major challenge for concurrent acquisition. Most artifact-removal strategies were developed for Cartesian, echo-planar imaging (EPI)–based fMRI, where repetitive gradient waveforms allow template subtraction or frequency-domain filtering. Such methods fail for non-Cartesian trajectories such as twisted projection imaging (TPI)[1], which feature smooth, continuously rotating gradients that generate distinct interference patterns. Sodium (^{23}Na) MRI, often using TPI for 3D acquisitions, faces additional difficulties due to its inherently low signal-to-noise ratio (SNR).

This study addresses gradient noise suppression in simultaneous ^{23}Na MRI–EEG using the 3D TPI sequence. The smooth TPI trajectory offers the potential to separate MRI-induced artifacts from neural activity more effectively than EPI. We compared three per-channel denoising methods: (1) Independent Component Analysis (ICA)[2], (2) a deep residual U-Net[3,4] trained to predict and subtract interference, and (3) a Variational Autoencoder (VAE)[5] that models gradient-noise distributions. EEG restoration quality was evaluated using power spectral density (PSD) analysis, and model performance was validated on ECG channels via root-mean-square error (RMSE) and Pearson correlation.

Methods

Participants and Acquisition: Twenty healthy volunteers (11 female, 9 male; 59.7 ± 19.5 years) participated under IRB approval. Each underwent a 90-min sodium MRI session on a 3 T Prisma scanner (Siemens, Erlangen, Germany) using a dual-tuned (^{1}H/^{23}Na) birdcage coil (QED, Cleveland, OH). Simultaneous EEG was recorded with a 32-channel Brain Vision system (Brain Products, Germany) while participants rested with eyes closed.

Each dataset comprised three to four 16-min TPI runs separated by 2-min no-scan intervals. EEG collected during no-scan periods served as clean reference, while scan-period EEG was labeled contaminated. Signals were band-pass filtered, resampled to 512 Hz, and segmented into 125 ms non-overlapping epochs.

Artifact Characterization: To isolate gradient interference, a watermelon phantom fitted with an EEG cap was scanned under identical TPI conditions (Figure 1d). Because phantom amplitudes differed from human data, an interference proxy was constructed by applying ICA to human EEG and selecting components whose normalized PSDs most closely matched phantom spectra, isolating MRI-related artifacts while excluding physiologic activity.

Denoising Models

ICA: Components matching interference spectra were removed before signal reconstruction.

Residual U-Net: A deep encoder–decoder with residual and skip connections was trained on synthetic data created by adding interference proxies to clean EEG segments:

$$x_{\text{cont}} = x_{\text{clean}} + x_{\text{int}} \xrightarrow{\text{Residual U-Net}} \hat{x}_{\text{int}} \implies \hat{x}_{\text{clean}} = x_{\text{cont}} - \hat{x}_{\text{int}} \approx x_{\text{clean}}$$

The network was trained on 384 samples and validated on 43, using single- and multi-channel inputs.

VAE: The VAE modeled the latent distribution of gradient noise and reconstructed artifacts for subtraction during inference.

Validation Using ECG: Because artifact-free EEG during scanning is unattainable, denoising accuracy was assessed on the ECG channel, where QRS complexes serve as stable references. Metrics included amplitude RMSE, z-scored RMSE, and Pearson correlation before and after denoising.

Results

The Residual U-Net provided the strongest artifact suppression and signal preservation, reproducing EEG spectra nearly identical to no-scan references while minimizing broadband and harmonic contamination. The single-channel model generalized best to in-vivo data, whereas the multi-channel variant excelled in synthetic tests where the interference proxy was well characterized (Figure 2). ICA recovered temporal structure but left high-frequency comb artifacts, and the VAE yielded smoother spectra but weaker physiologic fidelity (Figure 3). On ECG validation (Figure 4), the Residual U-Net reduced amplitude of RMSE by 54%, z-scored RMSE by 26%, and tripled Pearson correlation (Figure 5), confirming robust suppression without waveform distortion.

Discussion and Conclusion

Deep residual learning effectively denoises EEG recorded during non-Cartesian sodium MRI. The U-Net's hierarchical feature extraction and skip connections enable precise separation of smooth TPI gradient interference from neural activity, outperforming both ICA and VAE models. Consistent improvements across EEG and ECG metrics demonstrate strong cross-modality generalization. Although limited by synthetic training data, this framework provides a scalable foundation for real-time artifact suppression. Future extensions may incorporate self-supervised or attention-based architectures and larger datasets to improve robustness across scanners and sequences. Overall, the residual U-Net model substantially enhances EEG quality during ^{23}Na MRI, enabling more accurate characterization of neurophysiological and metabolic dynamics.

Acknowledgements

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Figures and Tables

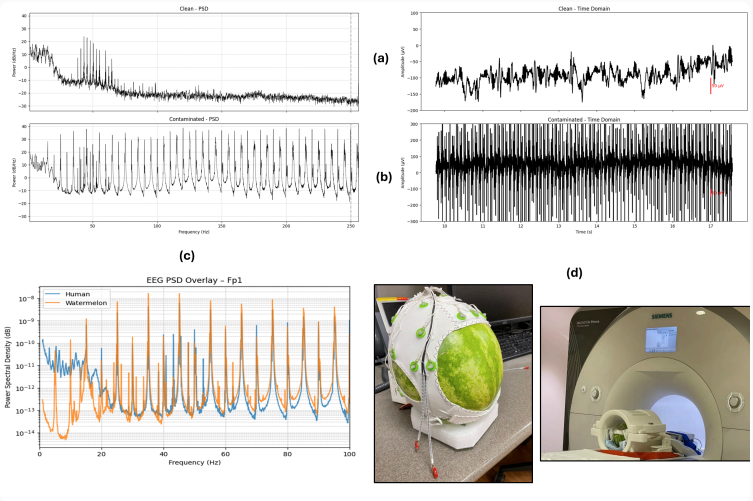

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Figure 1: *Characterization of MRI-induced EEG artifacts*
(a) Clean EEG segment showing normal low-frequency neural oscillations.
(b) EEG during MRI acquisition, exhibiting strong periodic interference and distortion from gradient switching.
(c) PSD comparison at electrode Fp1 between watermelon phantom and human EEG, revealing similar artifact periodicity with differing amplitudes.
(d) Watermelon phantom setup for interference-only acquisition inside the MRI bore and fitted with a 32-channel EEG cap.

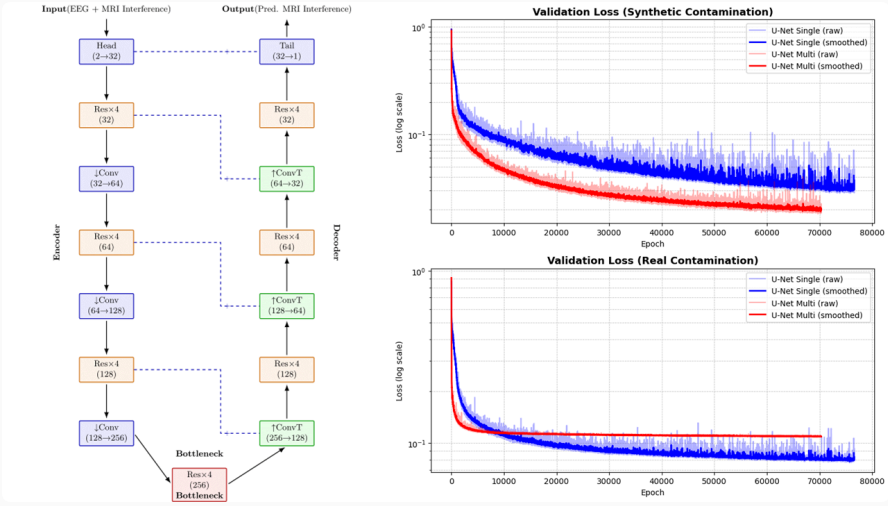

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Figure 2: *Architecture and Training of the Residual U-Net Model.*
Left: The encoder–decoder network structure with residual and skip connections that preserve temporal detail while suppressing gradient-induced artifacts.
Right: Validation loss curves for synthetic and real EEG contamination. The multi-channel model demonstrates faster and smoother loss convergence under synthetic contamination, whereas the single-channel variant achieves superior performance on real contamination.

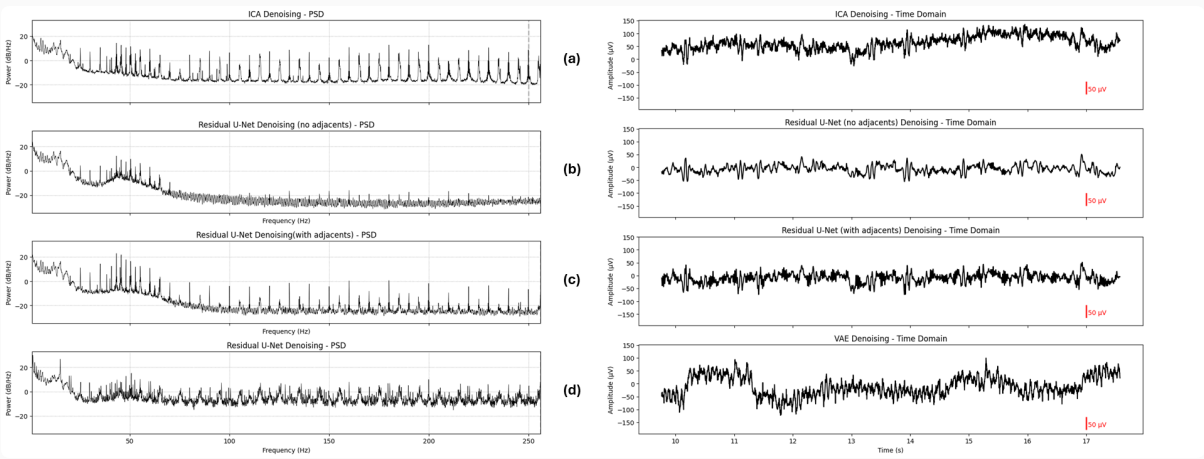

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Figure 3: *Spectra comparison of EEG denoising methods.*
(a) EEG after ICA artifact removal, showing partial recovery with residual high-frequency noise.
(b) EEG denoised via single-channel Residual U-Net, effectively restoring low-frequency neural activity.
(c) Multi-channel Residual U-Net result showing stronger artifact suppression through spatial context from neighboring electrodes.
(d) EEG processed by the VAE, yielding smoother spectra and improved temporal continuity relative to ICA

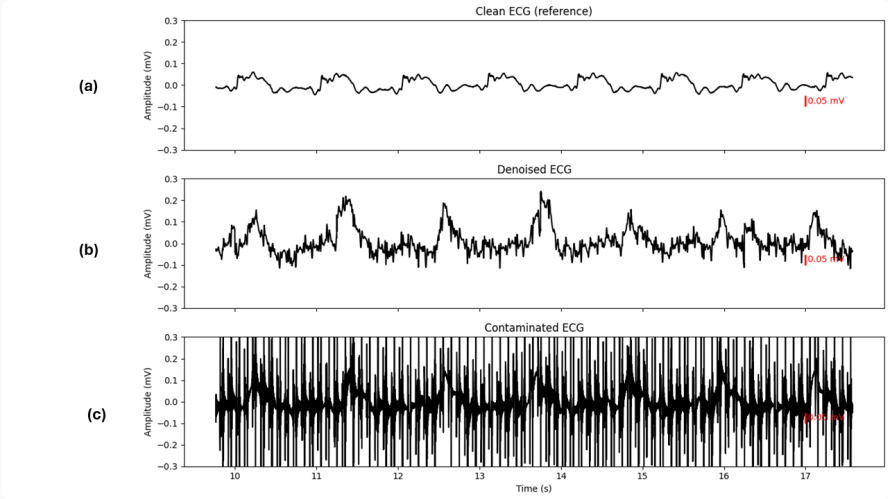

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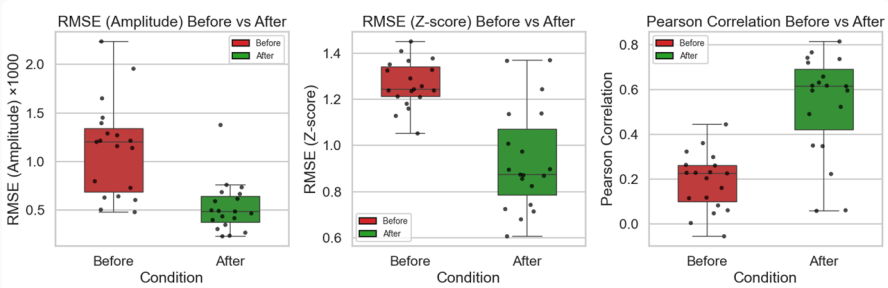
Figure 4: *Evaluation of ECG Denoising Performance.*
(a) Clean reference ECG recorded during no-scan periods. The U-Net model accurately restores waveform amplitude and timing, confirming robust artifact suppression and minimal distortion.
(b) ECG after U-Net–based denoising, showing recovery of QRS morphology and temporal alignment.
(c) Noise-contaminated ECG acquired during MRI, exhibiting large gradient-induced spikes.



Metric	Contaminated	Denoised	Mean Difference	p-value
RMSE (Amplitude)	0.00119 ± 0.00051	0.00056 ± 0.00027	-0.00063	< 0.001
RMSE (Z-scored)	1.273 ± 0.098	0.971 ± 0.258	-0.301	< 0.001
Pearson r (Amplitude)	0.184 ± 0.124	0.494 ± 0.281	+0.310	< 0.001

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Figure 5: *Quantitative and Visual evaluation of ECG denoising performance.*
Comparison of contaminated and denoised ECG traces relative to the clean reference across 20 subjects using paired t-tests. Boxplots show consistent improvements in amplitude and z-scored RMSE and Pearson correlation after denoising, indicating robust recovery of physiologic morphology. The table reports mean ± SD for each metric, with all $p < 0.001$, confirming effective artifact suppression and signal fidelity improvement.